

R-AODV: Reverse Ad-Hoc On-Demand Distance Vector Routing Algorithm With LoRa Mesh Based for Emergency Alert System

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Abstract—This research introduces a LoRa mesh network for emergency notification systems, which addresses the challenges of the conventional point-to-point LoRa system, such as limited transmission range and lack of backup paths. The mesh network comprises nodes that serve as relays, facilitating multi-hop data transmission to the gateway. The network also adapts to node failures and route changes by employing the Reverse Ad-hoc On-demand Distance Vector (R-AODV) routing algorithm. The performance of the proposed system was assessed through three types of tests: point-to-point, multi-hop, and node failure. The results indicate that the LoRa mesh network improves network coverage and reliability for emergency communication. The average delay for sending alert messages from the gateway neighbor node to the gateway is 0.82s, from the alarm node to the gateway is 1.54s in normal conditions, and 11.65s in case of a gateway neighbor node failure. The user satisfaction score is 4.31 out of 5.

Keywords— LoRa, Mesh, Point-to-Point, Emergency notification, AODV

I. INTRODUCTION

An emergency is an event that can occur unexpectedly at any time and can cause damage. Emergency notifications can be made through mobile phone networks, mobile applications, or communication devices such as walkie-talkies to reach relevant authorities. The purpose is to send alert signals to make those who can provide assistance aware of the situation and offer immediate help.

In situations where emergencies or disasters occur or in areas with poor mobile phone signal coverage and the absence of Walkie Talkie communication devices, there can be challenges in reporting emergencies to security guards. Consequently, this can delay the response and assistance efforts, potentially harming lives and property.

LoRa technology [1] is a wireless connectivity solution known for its long-range capabilities and low-power transmission. It offers robust anti-interference properties, making it suitable for large-scale deployments without the need for cable network infrastructure. Its applicability in emergency alert systems extends to various settings such as stadiums, exhibitions, and field hospitals. Moreover, the inherent capability of LoRa technology to form a mesh network enables the extension of signal coverage. Each node

selectively transmits signals to the nearest node, ensuring reliable communication. Even in the event of node damage, the network automatically establishes connections with neighboring nodes. This seamless operation facilitates the transmission of help requests over long distances, maximizing the chances of reaching security personnel promptly and enabling timely assistance.

In this paper, the researchers address the need for a robust routing algorithm to facilitate efficient and reliable emergency signal transmission within the LoRa Mesh network [4]. The proposed approach combines the utilization of LoRa Mesh communication technology with specialized algorithms designed to establish effective routes in ad hoc networks. The objective is to create a supportive system for emergency notifications, particularly in scenarios where the general communication network encounters issues or disruptions.

II. LITERATURE REVIEWS

A. Long-Range (LoRa)

LoRa (Long Range) is a wireless connectivity technology that is suitable for use in conjunction with Internet of Things (IoT) applications and direct device-to-device communication (Machine-to-Machine). It enables data transmission can be up to 15 kilometers. The connectivity format can be in the form of point-to-point (Fig.1), where devices communicate directly with each other without any intermediary devices. LoRa devices operate with low power consumption and can also be configured to enter power-saving modes when not actively transmitting data. This makes LoRa an energy-efficient solution, especially in scenarios where data transmission is infrequent.

LoRa operates in unlicensed frequency bands, which vary depending on the region. For example, in Europe, the frequency is typically 868 MHz, while in North America, it is 915 MHz. In Asia, including Thailand, the available frequency bands for LoRa are 433 MHz and 915 MHz, as specified in the IoT regulations of Thailand. These frequency bands provide the necessary spectrum for LoRa deployments, ensuring compatibility and adherence to local regulatory requirements for IoT applications in Thailand.

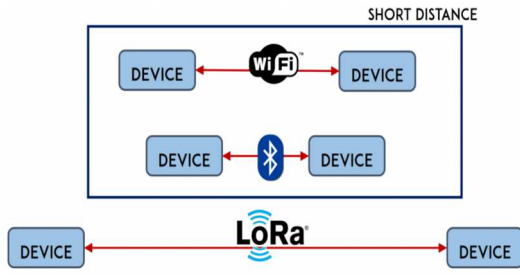


Fig. 1. LoRa point-to-point communication[2].

B. LoRa WAN

LoRaWAN (Long Range Wide Area Network) [3] shown in Fig.2, is a communications protocol for low-power wide area networks (LPWAN). LoRaWAN communications can enable IoT devices to connect over greater distances and consume less energy.

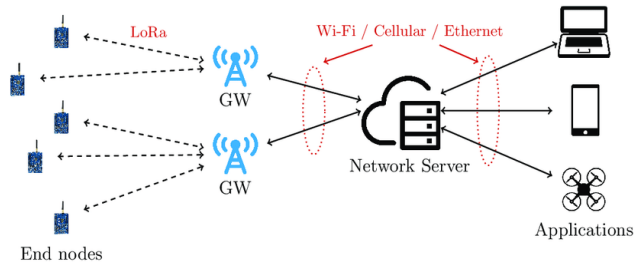


Fig. 2. LoRaWAN architecture diagram [3].

In a LoRaWAN architecture, the end node devices responsible for receiving or transmitting data typically include sensors for various measurements. They send data to gateways, which act as intermediaries for receiving and transmitting data between the source and the destination using frequency waves within the range of the transmitting devices. Once the data is received, the source node acts as a bridge, connecting the devices and the server. Data can be accessed and viewed through an application. Additionally, all data communication is encrypted for secure communications.

C. LoRa Mesh

LoRa Mesh is a network created using LoRa communication technology. The purpose of LoRa Mesh is to address the issue of signal coverage. Initially, Mesh networks were designed for military applications, hospitals, or large-scale areas requiring extensive communication connectivity.

In a Mesh network, each device serves as a node, communicating with other nodes within its range. Nodes are strategically placed to establish efficient communication with other nodes. Using dynamic routing, each node selects the optimal route to transmit signals to the destination node, considering factors like hop count, communication costs, and distance. Deploying the Mesh network effectively throughout an area enhances signal coverage and improves data transmission efficiency. Additionally, in the event of a node failure, the remaining nodes autonomously explore new routes to reach the destination. This autonomous behavior, self-discovery, enables nodes to find alternative paths for uninterrupted communication.

LoRa Mesh [4] technology offers significant advantages over LoRa point-to-point communication, providing extended coverage and cost savings. It operates independently of an

internet connection, making it adaptable to various environments. Nodes in the network automatically establish communication connections, eliminating the need for manual administration. The system's major limitation lies in the power supply to the devices, which can reduce node service time during high communication loads. However, this limitation is effectively mitigated with a primary focus on low-speed communication rather than image or video data. Overall, LoRa Mesh technology offers a flexible and efficient solution for wireless communication in challenging environments.

D. Ad hoc On-Demand Distance Vector (AODV) Routing Protocol

Ad hoc On-Demand Distance Vector (AODV) routing protocol [5] is a reactive protocol designed for use in mobile ad hoc networks (MANETs). It enables nodes to dynamically discover and maintain routes to other nodes in the network on an as-needed basis. AODV utilizes hop count as the decision metric for routing information, selecting the route that provides the minimum number of hop counts to the destination node. AODV uses route discovery and route reply messages to find the shortest path from the source node to the destination node, shown in Fig.3. The source node initiates route discovery by broadcasting a route request (RREQ) message with a path cost field. Each node receives the message updates the path cost, and re-broadcast. When the destination node receives a route request, it compares the 'path cost' field against previously received route request commands. If the path cost stored in the route request is better than any previously received, the destination node transmits a Route Reply (RREP) message back to the source node through the intermediate node.

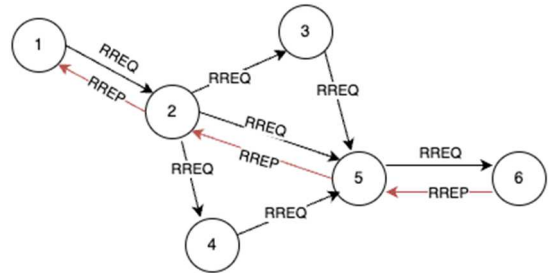


Fig. 3. AODV route discovery and route reply process [5].

E. LoRa and LoRa Mesh Applications

LoRa Mesh technology has been applied to reduce the maintenance costs of emergency power systems in office buildings [6], which can be costly and challenging to monitor. The solution integrates a LoRa wireless network with emergency lighting to enable collaboration and information exchange within the network. If one of the emergency lights is broken, the system can immediately notify the user, who can access and correct the issue. This helps reduce management costs. [7] developed a LoRa-based communication system, OWNGrid, for oil palm plantations. The system enables text and voice communication over long distances and low power consumption. The authors tested the system's functionality and performance in a plantation area and identified that the system's reliability needs to be improved. [8] presents a study on using LoRa technology to monitor the health of palm oil trees. The results show that the trunk and the canopy cause significant signal degradation and suggest that more research is needed to develop accurate path loss models for LoRa-IoT applications in palm-oil plantations. [9] developed an

emergency notification system using an ESP32 board and LoRa technology to address the problem of being stuck in a lift during an emergency and difficulty in requesting help due to mobile signal blockage by lift walls. The prototype device was developed in two models: the first model operates in point-to-point mode, consisting of a receiver and a transmitter, while the second version adds a signal repeater to increase the range of the emergency signal. Experimental results showed that the second model had 12-50% higher signal transmission efficiency than the first version and could be applied to real-life emergencies. The paper [10] compares the performance of ZigBee, LoRa, and IEEE1888 wireless communication technologies for community energy management systems. A hybrid system combining ZigBee and LoRa is proposed to balance high data rates and low power consumption. The results show that each technology has its strengths and can be used in different scenarios to achieve optimal performance. [11] explores the use of LoRa networks in Unmanned Aerial Vehicles (UAVs) for environmental monitoring. The method enables the rapid creation of large networks and is suitable for transmitting small amounts of data. The researchers designed a custom-slotted ALOHA Mesh protocol using LoRa and implemented custom software and hardware. The system uses multiple sensor nodes, including relays, gateways, and server networks. The results showed that with a minimized data rate of 0.293 kbps, the 1-hop transmission line exceeded 10 km, with a transmission delay of less than 6 seconds. The research [12] presents a protocol for using LoRa Mesh in a private network within an organization for IoT applications. The system supports communication between LoRa nodes and the LoRa gateway without the need for a service provider's LoRaWAN. However, the research has limitations, as it lacks clear test results and performance comparisons with other protocols. [13] presents a study on creating an efficient communication system using LoRa technology for emerging. The proposed Mesh network can transmit communication signals stably and exchange data between multiple nodes. Experimental results demonstrate the effectiveness of the proposed method in providing robust communication services in remote or disaster-prone areas.

The researchers propose to develop the LoRa technology used in the emergency notification system for passenger lifts at Prince of Songkla University, Phuket Campus [9]. The original system was unable to find a new route if the repeater is not working or if there is a problem with the service. In this paper, we propose to create a LoRa mesh network to increase the stability of the network service and the success rate of emergency notifications.

III. PROPOSED SYSTEM

A. System Architecture

In this paper, we present the design of an emergency notification system using LoRa Mesh technology. The system architecture consists of alarm nodes (node 2 and node 4), a gateway node, and gateway neighbor nodes (node 1 and node 3), as illustrated in Fig. 4; both use LoRa mesh connection technology to extend the network connection distance. Solid lines indicate the node connection in a range of signals, while dashed lines indicate that nodes are out of range. Each node connects and transmits data through other nodes, allowing the system to send out alerts even if one of the nodes is damaged. The node uses a route path from the proposed R-AODV (Reverse Ad-hoc On-demand Distance Vector) algorithm,

modified from the AODV routing algorithm. The gateway acts as a central point for the emergency notifications system. It broadcasts a list of source nodes to the alarm nodes that can be used to calculate the shortest path to the gateway.

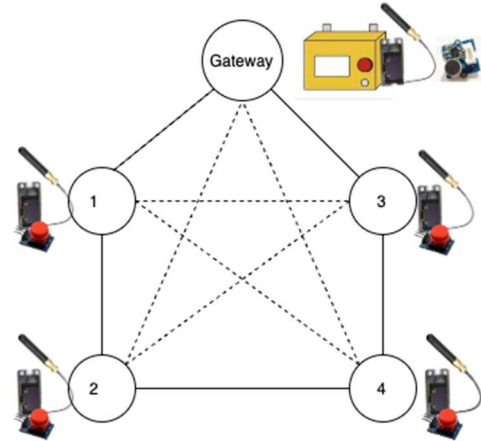


Fig. 4. System architecture.

B. Reverse Ad-hoc On-Demand Distance Vector (R-AODV) Routing Algorithm

The difference between AODV and R-AODV is that in AODV, nodes send route requests to find a path to the gateway themselves, while in R-AODV the gateway node broadcasts a beacon message with the gateway address and a blank node list. This message is modified and minimized based on the route advertisement message [14]. When a node receives the beacon message, it adds its node address to the node list. Each node broadcasts the received message to neighbor nodes. Eventually, all nodes in the LoRa mesh network use the node list data to define the best route path to the gateway and an alternative path, used in case the main path is unavailable (e.g., if a node becomes detached from the mesh). Additionally, all nodes periodically broadcast “ready” messages to announce their online status to neighboring nodes. This process is helpful in maintaining an active route from the alarm node to the gateway node through the active nodes. We design the R-AODV routing algorithm composed of three sub-algorithms running on a gateway, gateway neighbor node, and alarm node, illustrated in Fig. 5, 6, and 7.

1) *Gateway Node's Algorithm*: Fig. 5 shows that the gateway starts loading the default setting. It sends the message “R-AODV-Broadcast” to the surrounding nodes every 20 seconds. If there is an incoming message, it will check the message. For example, if the incoming message is “help,” it will contain the source address that sent the message and display the location on the monitor according to that address. And if an emergency response button is pressed, it will send the message “Gateway received” to the surrounding nodes.

2) *Alarm Node's Algorithm*: As shown in Fig. 6, the node starts by setting up the LoRa module and then sends an “R-AODV-Ready” message to its neighboring nodes every 10-14 seconds. If a message arrives, it checks what type of message it is. For example, if the message is an “R-AODV-Broadcast” message, it enters the R-AODV function to store the path and forward it to its neighbors. Then, it checks what

paths it has and selects the shortest one as the destination for sending alerts.

If the message is an “R-AODV-Ready” message, it stores the address of the neighboring node. If the message is a “Gateway received” message, a reply from the gateway, the node forwards the message to its neighbors. The node checks whether the address matches the receiver if the message is a “Help” message. If not, the node discards the message. If yes, the node forwards the message to the destination or gateway. When an alert is triggered, the node sends a “Help” message to the destination obtained from the R-AODV function. The sending will fail if the node does not have a destination in the system. The node status can be checked from the LED color. Red means not ready, green means ready.

The node also switches the LED every time it sends a message to indicate success and counts the number of sending rounds. When it reaches the predefined number of rounds, the node resets its values and updates its destination to ensure that the destination node in the system is still operational.

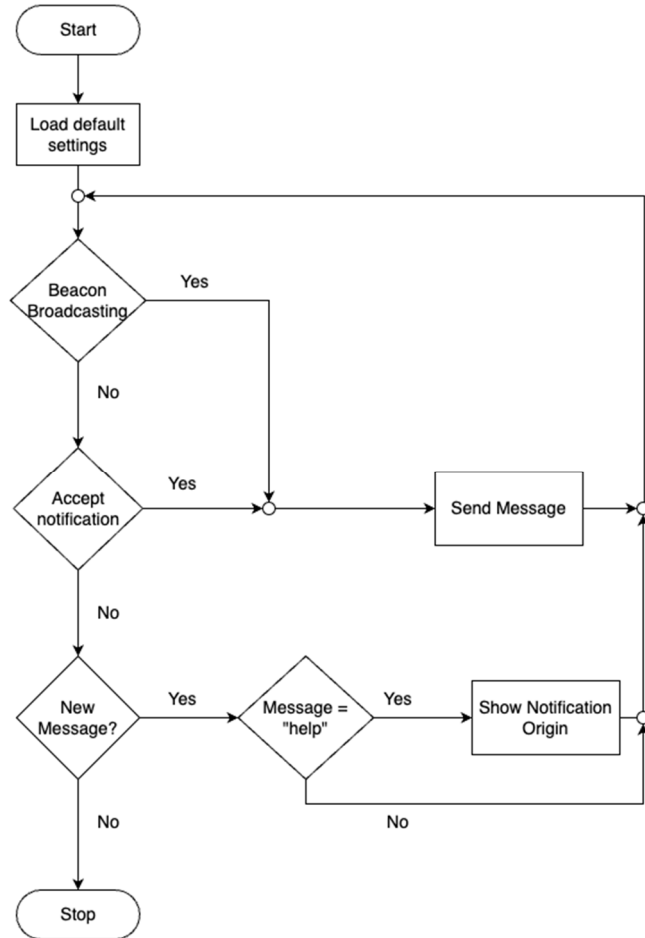


Fig. 5. Gateway node's algorithm.

3) *Gateway Neighbor Node's Algorithm:* As shown in Fig. 7, when a node connected to the gateway receives an “R-AODV Broadcast” message directly from the gateway, it stores the gateway address in its node list and forwards the message to its neighbors, excluding the gateway. The node will not update the gateway address if it receives a duplicate “R-AODV Broadcast” message.

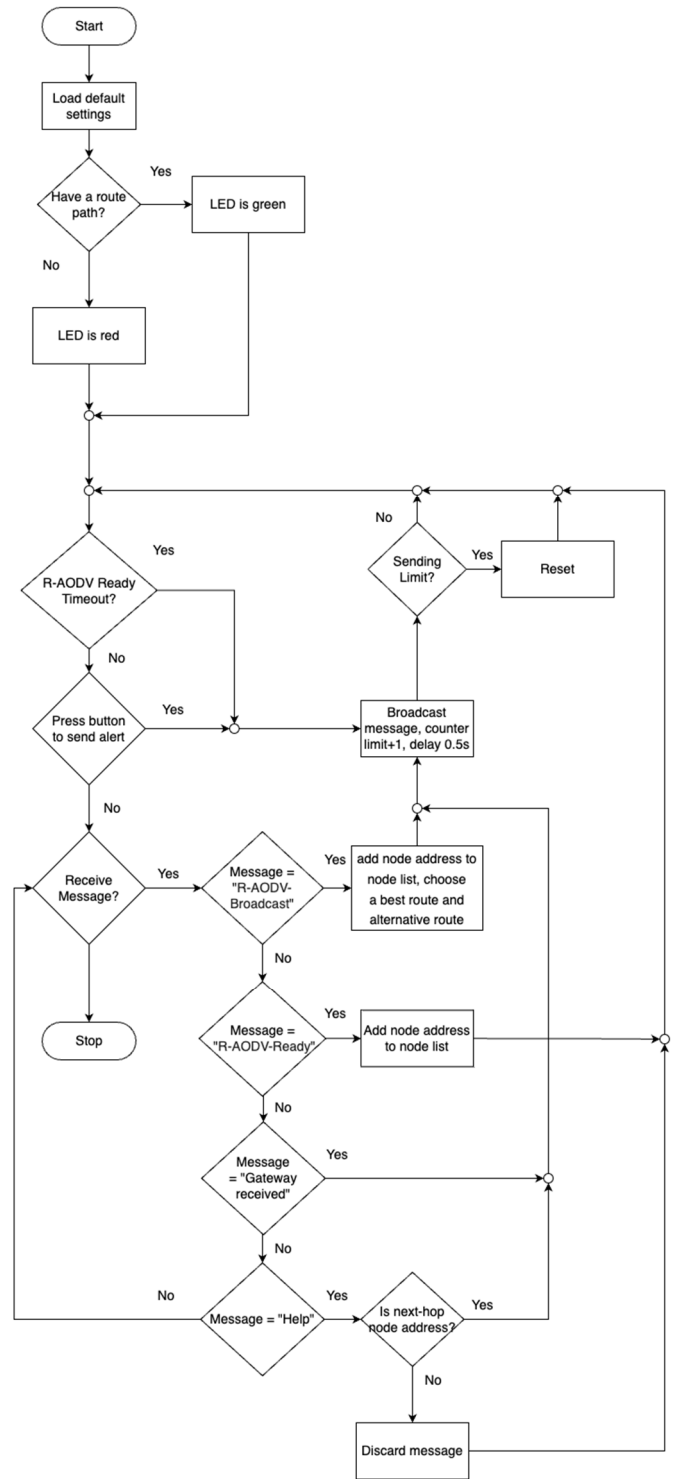


Fig. 6. Alarm node's algorithm.

C. Hardware Design

1) *Gateway device:* As shown in Fig. 8, the gateway device uses esp32acts as a receiver of alerts from the node that triggers them. It is designed as a compact rectangular box that is easy to install and has a function that displays information on an OLED screen and a buzzer sound when an alert is triggered by the device that sends the data. It has a button for responding to the help request. The device can be opened inside for convenient battery charging and maintenance. The functions of the device are all shown on the screen.

2) *Alarm/Node attached to Gateway device*: As shown in Fig. 9, the alert device resembles the gateway but has an LED instead of an OLED screen to reduce energy consumption and increase usage time. It sends data to the gateway for display as sound and text. The device's functions are listed in Table I.

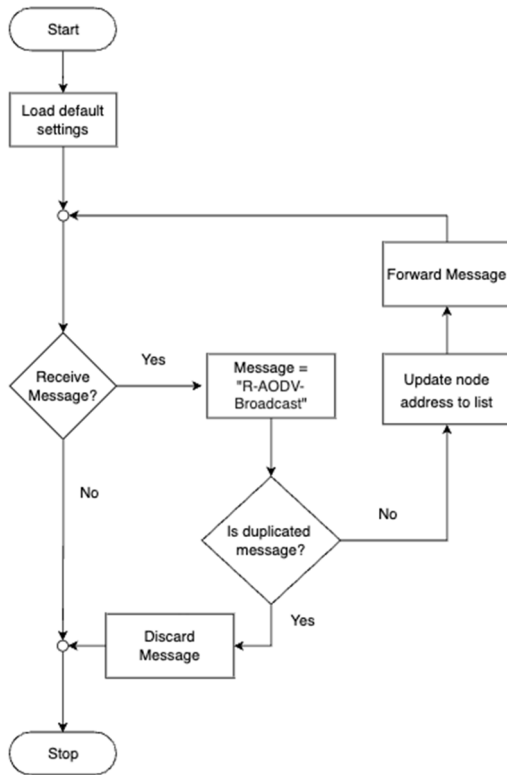


Fig. 7. Gateway neighbor node's algorithm.



Fig. 8. Gateway device.



Fig. 9. Alarm/Gateway neighbor node device.

TABLE I. DEVICE FUNCTIONS

Functions	Alarm/Node attached gateway ^a	Gateway
Alarm button	Yes	No
Display notification	No	Yes
LED status (Red/Green)	Yes	No
Sound alarm	Yes	No
Antenna	Yes	Yes
LoRa module	Yes	Yes

^a. The alarm device and the node connected to the gateway are interchangeable.

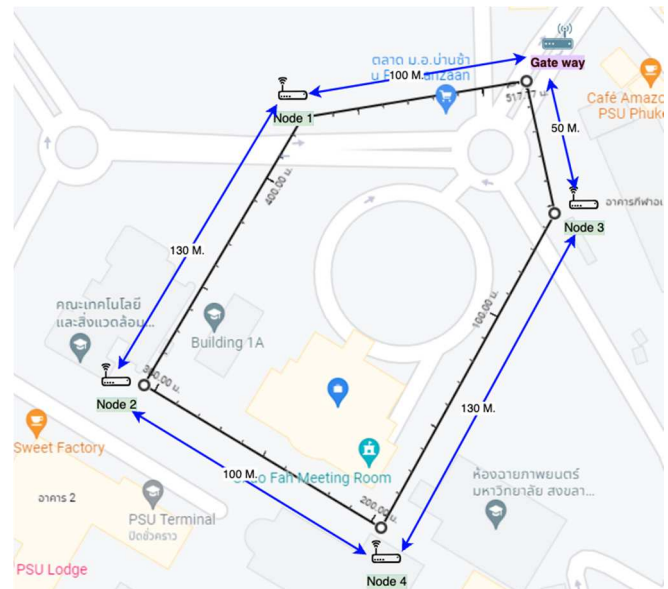


Fig. 10. Experiment setups (scenarios 1 to 3).

IV. RESULTS AND DISCUSSIONS

A. Experiment setups and results

The researchers designed and tested the system in three scenarios (20 times/scenario): Scenario 1, where the alert device triggers an alert from a node that receives a signal directly from the gateway and checks if it can send a signal to the gateway directly using the point-to-point technique. Scenario 2, where the alert device triggers an alert from a node that does not receive a signal from the gateway and checks if it can send a signal to the gateway using the multi-hop technique and select the shortest path among two possible paths to the gateway. Scenario 3, where a node stops working, checks if the node can use an alternative path to send a signal to the gateway if the main path fails. The average delay to send an emergency alert in scenarios 1, 2, and 3 are 0.82s, 1.54s, and 11.65s, respectively. Scenario 2 takes longer than Scenario 1 because the alarm node is not located next to the gateway, so the signal must be transmitted through other nodes. Scenario 3 takes the longest time because when some nodes stop working (node no. 1 or node no. 3), it takes time to find a new path. Therefore, sending a signal from the alarm node requires a new path, such as when node no.1 stops working, node no.2 sends an emergency alert through node no.4 to node no.3 and the gateway in sequence. The experiment setups and results are shown in Fig. 10 and Table II.

B. Evaluation of users satisfaction

The proposed prototype was tested by users at Prince of Songkla University, Phuket Campus. A total of 20 real users interested in using the system participated in the test. The system was tested one person at a time, with a focus on testing the emergency alert function. The test only involved triggering alerts from the alert device, not from the node attached to the gateway. The criteria for interpreting the average user satisfaction score are as follows: highest (4.51-5.00), high (3.51-4.50), moderate (2.51-3.50), low (1.51-2.50), and lowest (1.00-1.50). The average user satisfaction evaluation score was good at 4.31 out of 5, as shown in Table III.

TABLE II. EXPERIMENT RESULTS

Test cases	Setup	Result
Scenario 1: Point-to-Point (gateway and gateway neighbor node)	Press alert button from gateway neighbor node	Success (average delay = 0.82s, distance=50m or 100m)
Scenario 2: Multi-hop	Press from alarm node	Success, the alarm node chooses the shortest path to the gateway (average delay=1.54s, distance=180m or 230m)
Scenario 3: Turn off one of the gateway neighbor nodes ^b .	Press from alarm node while one of gateway neighbor node turn off	For instance, when node no.1 is turned off, the alert message transmission from node no.2 initially fails (with the current best route from node no.2 to the gateway) but eventually succeeds after the node is reset or delayed. This occurs because node no.2 waits for the timeout to expire before recalculating the optimal route (from node no.2 to node no.4, node no.3, and finally to the gateway). The average delay from the alarm node is 11.65s with a maximum distance of 230m/330m.

^b. Node no.1 and node no.3

TABLE III. EVALUATION OF USERS SATISFACTION

Function tests	Average	SD	Level of Satisfaction
Ease of use	3.5	2.0	Moderate
R-AODV algorithm	4.3	0.5	High
Success of alert system	4.5	0.5	High
LoRa Mesh	4.2	0.8	High
Be able to detect device status	4.5	0.5	High
Be able to change a route path	4.7	0.5	High
Practical benefits	4.5	0.5	High
Solving research problem	4.3	0.5	High

V. CONCLUSION

In this research, the LoRa Mesh technology was studied as a backup emergency alert system in case the existing system fails due to signal blockage or device malfunction. The devices were tested for point-to-point and multi-hop data transmission. The multi-hop experiment showed that clustering nodes increased the range and coverage of the network, outperforming point-to-point communication with its limited capacity and constraints. The R-AODV algorithm enhanced node routing, enabling efficient selection of the shortest and alternative paths. Future planning includes improving the R-AODV algorithm for excellent stability and supporting multiple gateways.

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